

CS & IT ENGINEERING



Computer Network

Routing

Lecture No. – 03



By - Abhishek Sir



Recap of Previous Lecture



Topic

Link State Routing

Topic

Distance Vector Routing



Topics to be Covered



Topic

Distance Vector Routing



ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

Telegram Link : https://t.me/abhisheksirCS_PW



[MCQ]

[GATE-IT-2007] [2 Mark]



#Q. For the network given in the figure below, the routing tables of the four nodes A, E, D and G are shown. Suppose that F has estimated its delay to its neighbors, A, E, D and G as 8, 10, 12 and 6 msec respectively and updates its routing table using distance vector routing technique.

$$F \rightarrow A = 8$$

$$F \rightarrow D = 12$$

$$F \rightarrow E = 10$$

$$F \rightarrow G = 6$$

Distance Vector

Routing Table of A	
A	0
B	40
C	14
D	17
E	21
F	9
G	24

Routing Table of D	
A	20
B	8
C	30
D	0
E	14
F	7
G	22

Routing Table of E	
A	24
B	27
C	7
D	20
E	0
F	11
G	22

Routing Table of G	
A	21
B	24
C	22
D	19
E	22
F	10
G	0

Routing Table of F		
Node	Cost	Via
A	8	A
B	20	D
C	17	E
D	12	D
E	10	E
F	0	F
G	6	G



Ans: A

A

A	8
B	<u>20</u>
C	<u>17</u>
D	12
E	10
F	<u>0</u>
G	6

~~**B**~~

A	<u>21</u>
B	8
C	7
D	19
E	14
F	<u>0</u>
G	22

~~**C**~~

A	8
B	20
C	17
D	12
E	10
F	<u>16</u>
G	6

~~**D**~~

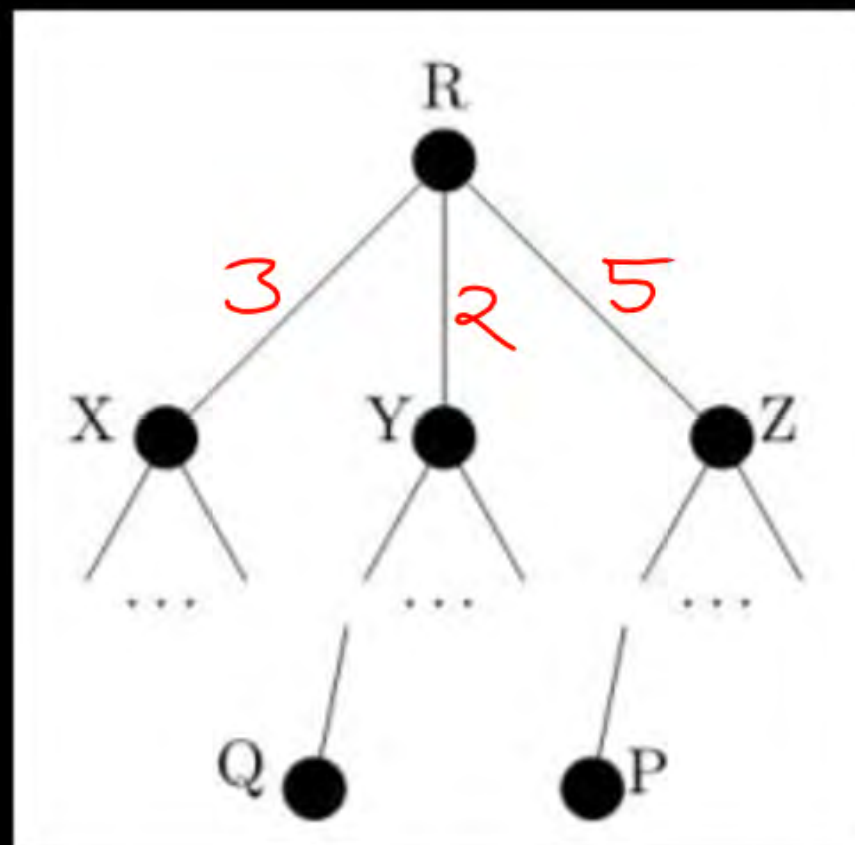
A	8
B	<u>8</u>
C	7
D	12
E	10
F	<u>0</u>
G	6

[MSQ]

[GATE-2021][2 Mark]



#Q. Consider a computer network using the distance vector routing algorithm in its network layer. The partial topology of the network is shown below.



The objective is to find the shortest-cost path from the router R to routers P and Q. Assume that R does not initially know the shortest routes to P and Q. Assume that R has three neighbouring routers denoted as X, Y and Z.

During one iteration, R measures its distance to its neighbours X, Y, and Z as 3, 2 and 5, respectively. Router R gets routing vectors from its neighbours that indicate that the distance to router P from routers X, Y and Z are 7, 6 and 5, respectively. The routing vector also indicates that the distance to router Q from routers X, Y and Z are 4, 6 and 8 respectively. Which of the following statement(s) is/are correct with respect to the new routing table of R, after updation during this iteration?

- ☒ A The distance from R to P will be stored as 10 F
- ☒ B The distance from R to Q will be stored as 7 T
- ☒ C The ^(via) next hop router for a packet from R to P is Y T
- ☒ D The next hop router for a packet from R to Q is Z F

Ans: B & C

$$C(\underline{R}, \underline{X}) = \underline{3},$$

$$C(\underline{R}, \underline{Y}) = \underline{2},$$

$$C(\underline{R}, \underline{Z}) = \underline{5},$$

$$\underline{X} : D_X(P) = \underline{7} \text{ and}$$

$$D_X(Q) = 4$$

$$\underline{Y} : D_Y(P) = \underline{6} \text{ and}$$

$$D_Y(Q) = 6$$

$$\underline{Z} : D_Z(P) = \underline{5} \text{ and}$$

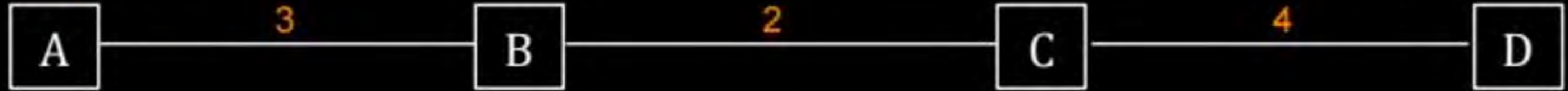
$$D_Z(Q) = 8$$

$$\underline{R} : \underline{D_R(P)} = ? \text{ and } \underline{D_R(Q)} = ?$$

$$\begin{aligned} D_R(P) &= \min[\{C(R, X) + D_X(P)\}, \{C(R, Y) + D_Y(P)\}, \{C(R, Z) + D_Z(P)\}] \\ &= \underline{8} (\text{Via } \underline{Y}) = \min[(3+7), (2+6), (5+5)] \end{aligned}$$

$$\begin{aligned} D_R(Q) &= \min[\{C(R, X) + D_X(Q)\}, \{C(R, Y) + D_Y(Q)\}, \{C(R, Z) + D_Z(Q)\}] \\ &= \underline{7} (\text{Via } \underline{X}) = \min[(3+4), (2+6), (5+8)] \end{aligned}$$

Intialization :-



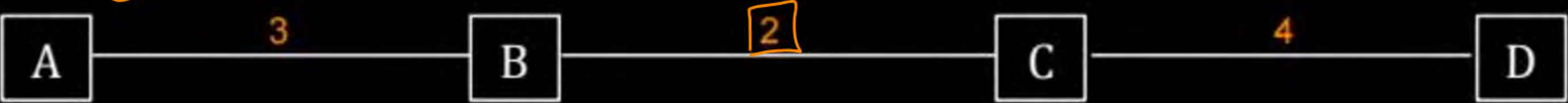
Node	Cost	Via
A	0	A
B	3	B

Node	Cost	Via
A	3	A
B	0	B
C	2	C

Node	Cost	Via
B	2	B
C	0	C
D	4	D

Node	Cost	Via
C	4	C
D	0	D

First Round of Update :-

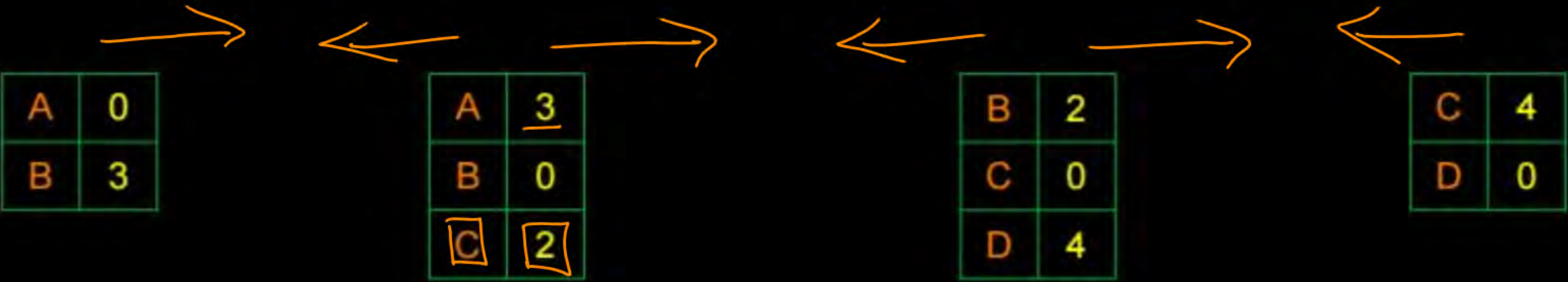


Node	Cost	Via
A	<u>0</u>	A
B	<u>3</u>	<u>B</u>

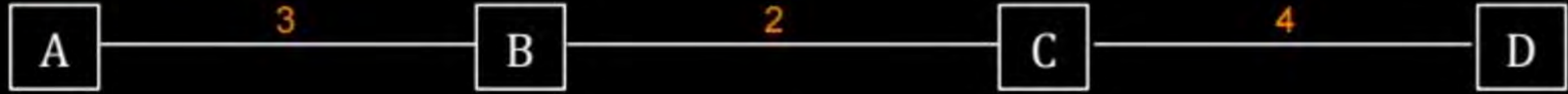
Node	Cost	Via
A	<u>3</u>	A
B	<u>0</u>	<u>B</u>
C	<u>2</u>	<u>C</u>

Node	Cost	Via
B	<u>2</u>	<u>B</u>
C	<u>0</u>	<u>C</u>
D	<u>4</u>	<u>D</u>

Node	Cost	Via
C	<u>4</u>	C
D	<u>0</u>	<u>D</u>



After First Round :-



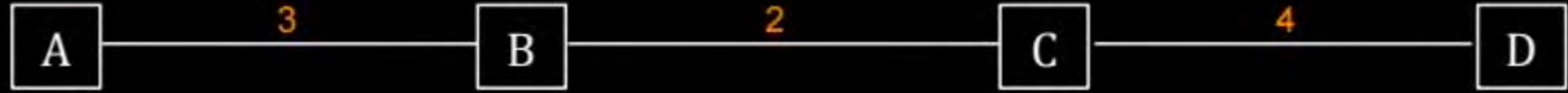
Node	Cost	Via
A	0	A
B	3	B
C	5	B

Node	Cost	Via
A	<u>3</u>	<u>A</u>
B	<u>0</u>	<u>B</u>
C	<u>2</u>	<u>C</u>
D	6	C

Node	Cost	Via
A	5	B
B	<u>2</u>	<u>B</u>
<u>C</u>	<u>0</u>	<u>C</u>
D	<u>4</u>	<u>D</u>

Node	Cost	Via
B	6	C
C	<u>4</u>	<u>C</u>
<u>D</u>	<u>0</u>	<u>D</u>

Second Round of Update :-



Node	Cost	Via
<u>A</u>	<u>0</u>	A
B	3	B
C	5	B

Node	Cost	Via
A	3	A
B	0	B
C	2	C
D	6	C

Node	Cost	Via
A	5	B
B	2	B
C	0	C
D	4	D

Node	Cost	Via
<u>B</u>	<u>6</u>	C
C	4	C
D	0	D

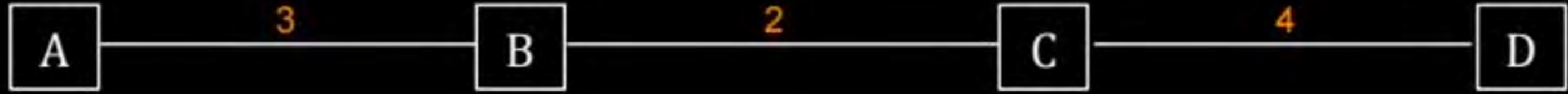
A	0
B	3
C	5

A	3
B	0
C	2
D	6

A	5
B	2
C	0
D	4

B	6
C	4
D	0

After Second Round :-



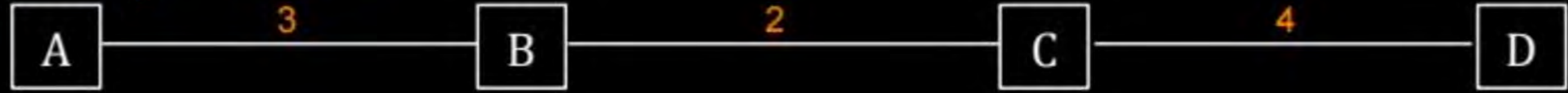
Node	Cost	Via
<u>A</u>	<u>0</u>	<u>A</u>
<u>B</u>	<u>3</u>	<u>B</u>
<u>C</u>	<u>5</u>	<u>B</u>
<u>D</u>	<u>9</u>	<u>B</u>

Node	Cost	Via
<u>A</u>	<u>3</u>	<u>A</u>
<u>B</u>	<u>0</u>	<u>B</u>
<u>C</u>	<u>2</u>	<u>C</u>
<u>D</u>	<u>6</u>	<u>C</u>

Node	Cost	Via
<u>A</u>	<u>5</u>	<u>B</u>
<u>B</u>	<u>2</u>	<u>B</u>
<u>C</u>	<u>0</u>	<u>C</u>
<u>D</u>	<u>4</u>	<u>D</u>

Node	Cost	Via
<u>A</u>	<u>9</u>	<u>C</u>
<u>B</u>	<u>6</u>	<u>C</u>
<u>C</u>	<u>4</u>	<u>C</u>
<u>D</u>	<u>0</u>	<u>D</u>

Third Round of Update :-



Node	Cost	Via
A	0	A
B	3	B
C	5	B
D	9	B

Node	Cost	Via
A	<u>3</u>	<u>A</u>
B	<u>0</u>	B
C	<u>2</u>	<u>C</u>
D	<u>6</u>	<u>C</u>

Node	Cost	Via
A	5	B
B	2	B
C	<u>0</u>	C
D	<u>4</u>	<u>D</u>

Node	Cost	Via
A	9	C
B	6	C
C	4	C
D	0	D

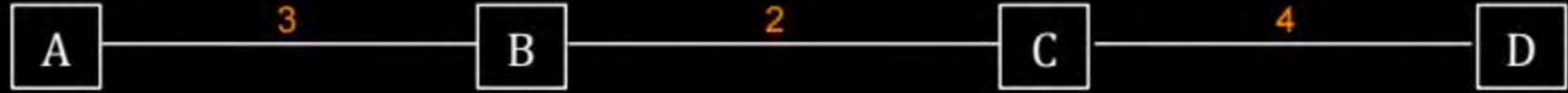


A	0
B	3
C	5
D	9



A	9
B	6
C	4
D	0

After Third Round:-



Node	Cost	Via
A	0	A
B	3	B
C	5	B
D	9	B

Node	Cost	Via
A	3	A
B	0	B
C	2	C
D	6	C

Node	Cost	Via
A	5	B
B	2	B
C	0	C
D	4	D

Node	Cost	Via
A	9	C
B	6	C
C	4	C
D	0	D



Topic : Distance Vector Routing



* Asynchronous

* Synchronous
↓
RIP

=> Distributed : (Decentralized)

- Each node notifies neighbors only when its "Distance Vector" changes
- Neighbors then notify their neighbors only, if necessary
- No notification received, no action taken

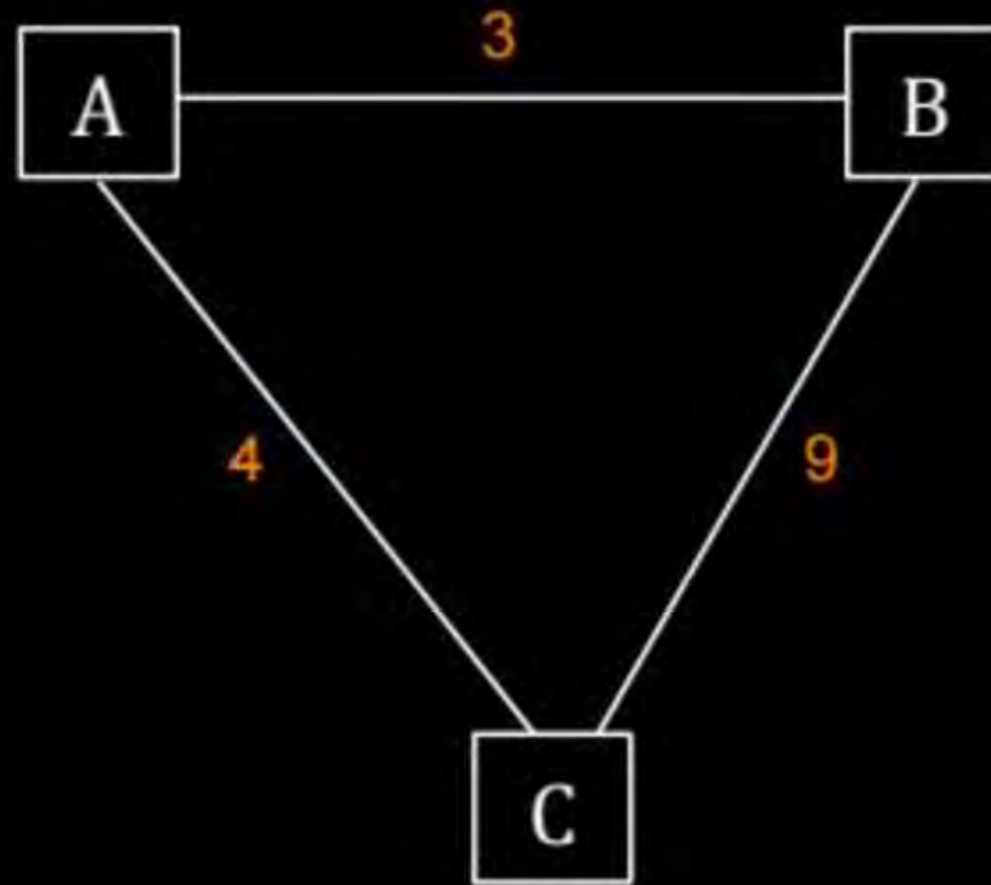
=> Iterative :

- Process iterate until no new information
[Available to be exchanged between neighbors]

=> Asynchronous

Initialization :-

Node	Cost	Via
A	0	A
B	3	B
C	4	C



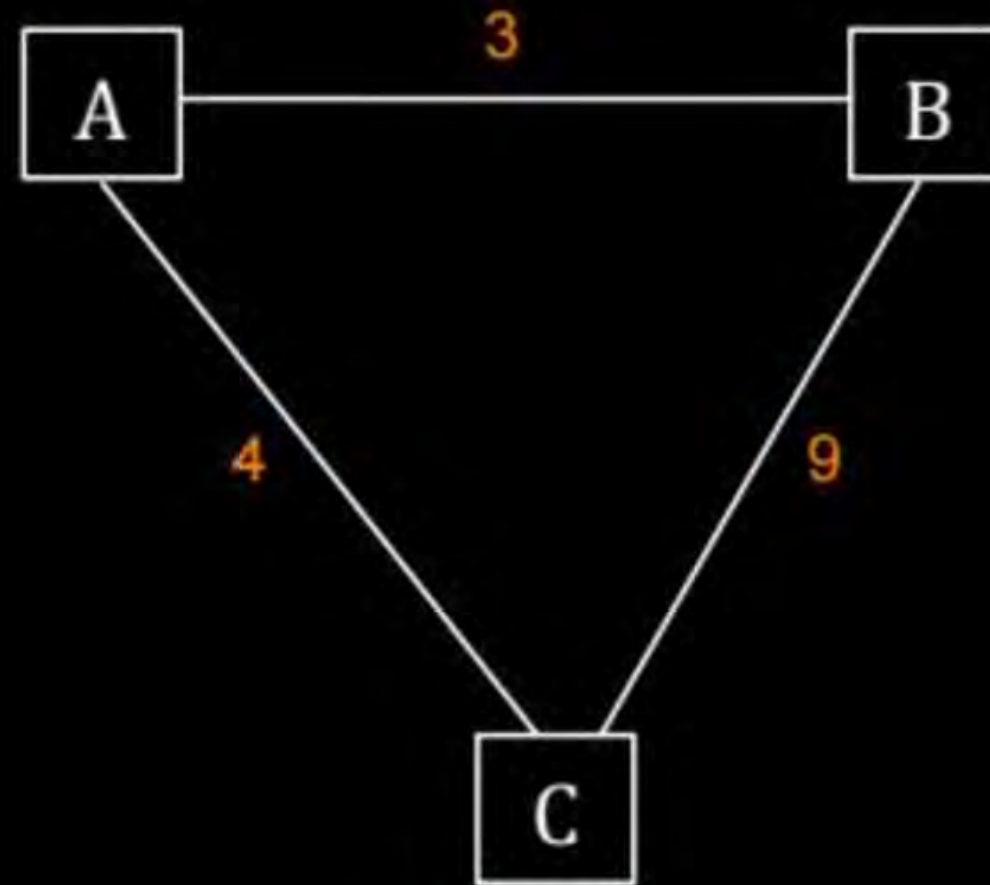
Node	Cost	Via
A	3	A
B	0	B
C	9	C

Node	Cost	Via
A	4	A
B	9	B
C	0	C

After First Round :-



Node	Cost	Via
A	<u>0</u>	<u>A</u>
B	3	B
C	4	C



Node	Cost	Via
A	3	A
B	<u>0</u>	<u>B</u>
C	7	A

*

Node	Cost	Via
A	4	A
B	7	A
C	<u>0</u>	<u>C</u>

*



Topic : Distance Vector Routing

* Asynchronous



→ Each router wait for
either “changes in adjacent link cost”
or “receives new “Distance Vector” estimate from any neighbor”

→ It update its own “Distance Vector” estimate using Bellman-Ford equation

→ if any update in its own “Distance Vector” estimate
then send own updated “Distance Vector” estimate to their neighbors
else “do nothing”

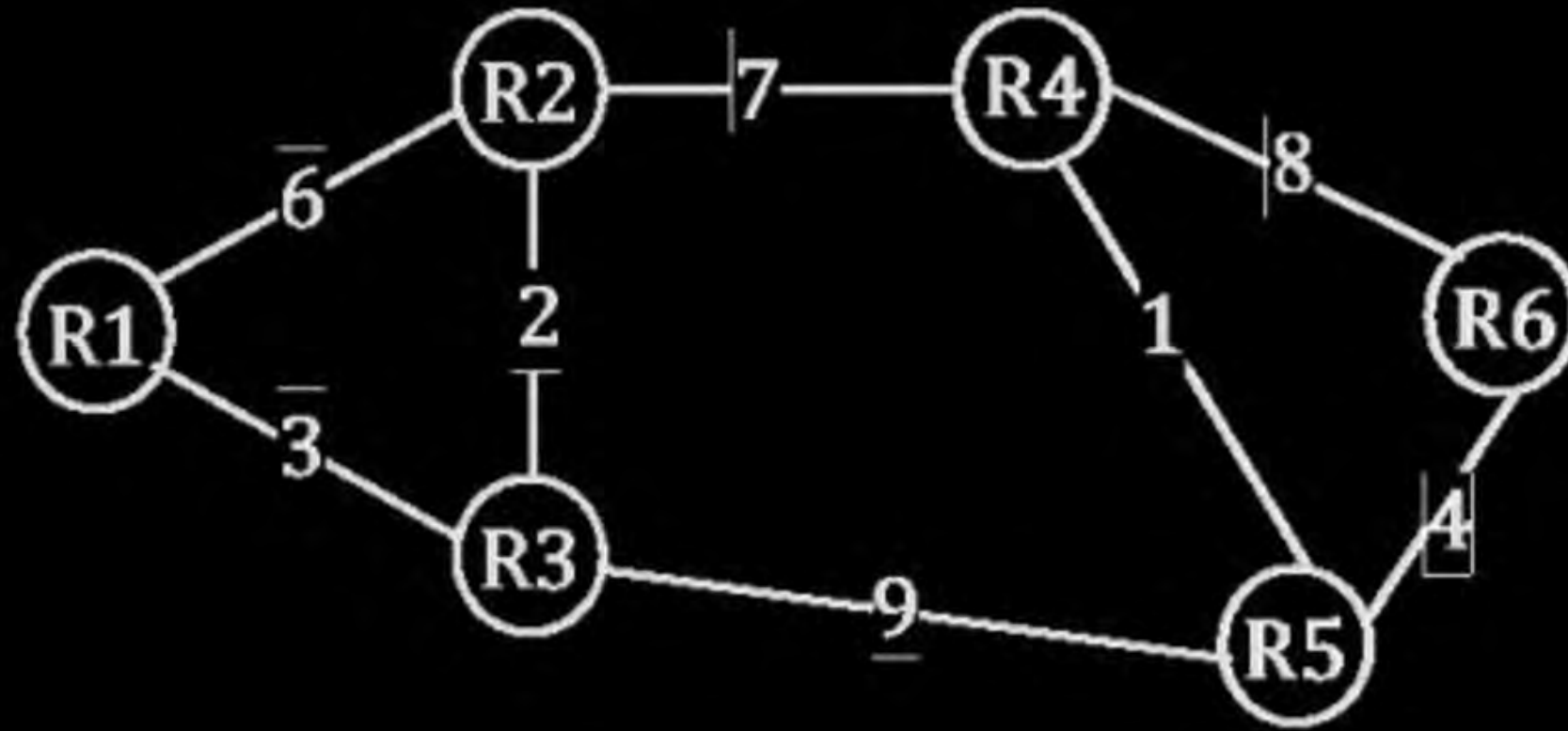
[MCQ]

H.W. IIT-G

[GATE-2010][2 Mark]



#Q. Consider a network with 6 routers R1 to R6 connected with links having weights as shown in the following diagram :



All the routers use the distance vector based routing algorithm to update their routing tables. Each router starts with its routing table initialized to contain an entry for each neighbor with the weight of the respective connecting link. After all the routing tables stabilize, how many links in the network will never be used for carrying any data ?

- A 1
- B 2
- C 3
- D 4

[MCQ]

Linked ques. (H.W.)

[GATE-2010][2 Mark]



#Q. Suppose the weights of all unused links in the previous question are changed to 2 and the distance vector algorithm is used again until all routing tables stabilize. How many links will now remain unused ?

- A 0
- B 1
- C 2
- D 3

Intialization :-



Node	Cost	Via
A	0	A
B	1	B

Node	Cost	Via
A	1	A
B	0	B
C	1	C

Node	Cost	Via
B	1	B
C	0	C

First Round of Update :-



Node	Cost	Via
A	0	A
B	1	B

Node	Cost	Via
A	1	A
B	0	B
C	1	C

Node	Cost	Via
B	1	B
C	0	C

A	0
B	1

A	1
B	0
C	1

B	1
C	0

After First Round :-



Node	Cost	Via
A	0	A
B	1	B
C	2	B

Node	Cost	Via
A	1	A
B	0	B
C	1	C

Node	Cost	Via
A	2	A
B	1	B
C	0	C

Second Round of Update :-



Node	Cost	Via
A	0	A
B	1	B
C	2	B

Node	Cost	Via
A	1	A
B	0	B
C	1	C

Node	Cost	Via
A	2	B
B	1	B
C	0	C



A	0
B	1
C	2



A	2
B	1
C	0

After Second Round :-



Node	Cost	Via
A	0	A
B	1	B
C	2	B

Node	Cost	Via
A	1	A
B	0	B
C	1	C

Node	Cost	Via
A	2	B
B	1	B
C	0	C

CASE I (First Round):-



Node	Cost	Via
A	0	A
B	∞	
C	2	B

Node	Cost	Via
A	∞	
B	0	B
C	1	C

Node	Cost	Via
A	2	B
B	1	B
C	0	C



A	∞
B	0
C	1

CASE I (Second Round):-



Node	Cost	Via
A	0	A
B	∞	
C	∞	

Node	Cost	Via
A	∞	
B	<u>0</u>	<u>B</u>
C	<u>1</u>	<u>C</u>

Node	Cost	Via
A	<u>∞</u>	
B	<u>1</u>	<u>B</u>
C	<u>0</u>	<u>C</u>



A	∞
B	1
C	0



2 mins Summary



Topic

Distance Vector Routing →



THANK - YOU

